6. If the size of sample is larger than 30, the t-distribution tends to
   a) normal distribution b) F-distribution
   c) chi-square distribution d) Poisson distribution

7. If a random sample of 10 observations has variance 324 then standard error is
   a) $18/\sqrt{10}$ b) $18/10$
   c) $10/18$ d) $2/\sqrt{5}$

8. A sample of 16 units was taken for testing an hypothesis concerning the mean of a normal population. Then the degrees of freedom of the appropriate test statictic is
   a) 14 b) 15
   c) 16 d) 8

9. If s_1^2 and s_2^2 are respectively the variance of two independent random samples of sizes 'm' and 'n'. Then standard deviation of the combined sample is
   a) $\sqrt{\frac{ms_1^2 + ns_2^2}{m+n}}$ b) $\sqrt{\frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n}}$
   c) $\sqrt{\frac{ms_1^2 + ns_1^2}{m+n+2}}$ d) $\sqrt{\frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2}}$

10. A company gave an intensive training to its salesman to increase the sales. A random sample of 6 salesmen was selected and the value of their sales made before and after the training is recorded. Which test will be more appropriate to test whether there is an increase in mean sales
    a) normal test b) paired t-test c) χ^2-test d) F-test

11. If the order of the contigency table is (5×4). Then the degree of freedom of the corresponding chi-square test statistic is
    a) 18 b) 17 c) 12 d) 25

12. For testing the hypothesis concerning variance of a normal population _____ is used.
    a) t-test b) F-test c) Z-test d) χ^2-test

13. If σ^2 is the variance of normal population, then the degrees of freedom of the sampling distribution of the test statistic for testing $H_0 : \sigma^2 = \sigma_0^2$ is:
    a) $n-1$ b) $n+1$ c) n d) $n-2$

14. If n is the degree of freedom of chi-square distribution then its variance is
    a) n b) $n-1$ c) $2n$ d) $n+1$

15. If chi-square is performed for testing goodness of fit to a data with k classes on estimating 's' parameters then degrees of fredom of test statistic is.
    a) $k-s$ b) $(k-1)(s-1)$
    c) $k-1-s$ d) $k-1$

16. The statistic χ^2, with usual notations, in case of contingency table of order $(m \times n)$ is given by
    a) $\chi_0^2 = \sum_{i=1}^{m}\sum_{j=1}^{n} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$ b) $\chi_0^2 = \sum_{i=1}^{k}\left[\frac{(O_i - E_i)}{E_i}\right]^2$
    c) $\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)}{E_i}$ d) $\chi_c^2 = \sum_{i=1}^{k} \frac{O_i}{E_i}$

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